KneeGait-Edge: Technical Code Review Specification

Designated Reviewer: **Rahul** (Team Leader & AI / Kinematic Analytics Lead) | Hackathon: ZINNOVATIO 4.0

Project: KneeGait-Edge Wearable	**Target Hardware:** ESP32 Dual-IMU (BLE 50Hz)
Scope: Real-Time BLE Streaming Implementation	**Date:** October 30, 2026
Status: Ready for Review & Sign-off	**Validation Base:** Bytyqi (2014) & Rhudy (2024)

This specification outlines the exact codebase files implemented to replace the synthetic browser simulation with a direct, persistent **50 Hz Web Bluetooth (BLE)** streaming pipeline. As the Kinematic Analytics Lead, Rahul's role is to verify the clinical validity of the real-time gait segmentation heuristics, discrete biomarker extraction calculations, and presentation benchmarks.

1. Primary Signal Parsing & Gait Phase Segmentation

File: `src/lib/bleService.ts` | **Priority:** **HIGH (CRITICAL)**

This file establishes the native Web Bluetooth GATT connection with the ESP32 peripheral (Service: `4fafc201-1fb5-459e-8fcc-c5c9c331914b`, Char: `beb5483e-36e1-4688-b7f5-ea07361b26a8`). The critical section requiring algorithmic sign-off is the dynamic stance-versus-swing phase classifier.

```
// Lines 150-165 in src/lib/bleService.ts
if (kneeAngle > 20 || (kneeAngle > 12 && angularVel > 25)) {
    this.currentPhase = 'swing';
} else if (kneeAngle <= 14 && angularVel < 5) {
    this.currentPhase = 'stance';
}
```

Review Checklist for Rahul:
• Does the 20° flexion threshold cleanly capture foot toe-off across both normal and stiff gait?
• Is the angular velocity threshold ($\omega > 25$°/s) sufficiently robust against ground impact tremors?
• Confirm that hysteresis between 14° and 20° prevents phase chattering during weight transition.

2. Biomechanical Feature Extraction & Cadence Engine

File: `src/context/BleContext.tsx` | **Priority:** **HIGH (CRITICAL)**

Maintains persistent streaming across all frontend navigation routes. Manages the circular 8-second waveform buffer (400 samples @ 50 Hz) and computes gait metrics upon detected heel-strikes.

```
// Peak Swing Flexion Tracking & Heel-Strike Cadence Trigger
if (sample.phase === 'swing') {
    swingMaxRef.current = Math.max(swingMaxRef.current, sample.knee);
}
if (prevPhaseRef.current === 'swing' && sample.phase === 'stance') {
    lastPeakRef.current = swingMaxRef.current;
    swingMaxRef.current = 0;
    if (lastStrikeRef.current !== null && sample.t > lastStrikeRef.current) {
        const dt = sample.t - lastStrikeRef.current;
        if (dt > 0.4 && dt < 2.5) {
            cadenceRef.current = Math.round(120 / dt);
        }
    }
    lastStrikeRef.current = sample.t;
}
```

Review Checklist for Rahul:

• Verify stride validity window: 0.4s < Δt < 2.5s (corresponds to physiological 48–300 steps/min).
• Validate Cadence formula: *Cadence = 120 / Δt_stride* (calculates 2 steps per complete gait cycle).
• Verify sliding Range of Motion (ROM): *ROM = max(buffer) - min(buffer)* over 8.0s visible window.

3. Dashboard & Clinical Decision Boundaries

Files: `src/pages/Dashboard.tsx` & `src/components/RiskBadge.tsx` | **Priority:** **MEDIUM**

Replaced mock patient buttons ('Healthy gait' / 'Stiff gait') with live BLE connection indicators and connected live peak values to the clinical decision rules established by Bytyqi et al. (2014):

Peak Swing Range	Clinical Classification	UI Badge Response
θ_peak < 45.0°	Medial Compartment KOA Stiff-Gait Strategy	■ **Stiff Gait Risk (Alert)**
45.0° ≤ θ_peak < 49.4°	Borderline / Mild Extension Restriction	■ **Borderline Gait**
49.4° ≤ θ_peak ≤ 65.0°	Normal Physiological Sagittal Kinematics	■ **Normal Gait (Clear)**

4. Mathematical Blueprint & Pitch Documentation

File: `KINEMATIC_PIPELINE_ARCHITECTURE.md` | **Priority:** **DOCUMENTATION / PITCH LEAD**

A 9-section master reference detailing the complementary filter derivation (α = 0.96 at 50 Hz), soft-tissue artifact (STA) suppression on the flat anteromedial tibia, BLE GATT packet structure, and Random Forest classifier feature weights. Rahul should review this file prior to the final hackathon pitch.

REVIEW & APPROVAL SIGN-OFF	DATE & VERDICT
Rahul (Team Leader & AI / Kinematic Analytics Lead) I have examined the signal segmentation thresholds, feature extraction math, and clinical decision boundaries implemented in the BLE streaming pipeline.	Date: _____ Status: [ ] APPROVED [ ] REVISION REQ.